

# Basics of University Mathematics

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# Linear Systems of Equations with No Solution

Let us first recall what this case looked like when solving pairs of equations.

## Basic Exercise

Solve the following systems of equations with two variables both algebraically and graphically:

$$(a) \begin{cases} y = x + 1 \\ 2(y - x) = 4 \end{cases} \qquad (b) \begin{cases} y = -x + 2 \\ y = x \\ y = -x \end{cases}.$$

When the equations of a pair of equations with two variables are combined such that the variable  $y$  (or  $x$ ) is eliminated, it is required that the lines corresponding to the equations are at the same height (width). Thus, from the resulting equation with one variable, the points on the  $x$ -axis (or  $y$ -axis) can be solved where this is the case.

In the previous task (a), proceeding in this way leads to the false equation  $0 = 2$ . This means that there are no such values for the variable  $x$ , or that the lines never intersect. Therefore, the system of equations has no solution. If the task is solved using the Gauss algorithm, the same thing is observed from the lowest false row of the augmented matrix

$$\left[ \begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 0 & 2 \end{array} \right]$$

which corresponds to the equation  $0 = 2$ .

Although the lines of the equation system in part (b) intersect, they do not all intersect at the same point. Thus, the system of equations has no solution, and the attempt to solve it ends with the false equation:

$$\left[ \begin{array}{cc|c} 1 & 1 & 2 \\ -1 & 1 & 0 \\ 1 & 1 & 0 \end{array} \right] \Longleftrightarrow \left[ \begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 2 & 2 \\ 0 & 0 & -2 \end{array} \right] \Longleftrightarrow \left[ \begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{array} \right].$$

In this case, the first two rows give the intersection point of the first two equations, but since the third equation is not true, the aforementioned intersection point does not satisfy the third equation, so the system of equations has no solution.

Similarly, deriving a solution for a system of equations with three or more variables leading to a false equation always means that the system of equations has no solution. What does this mean geometrically in the case of three variables and (a) two (b) three equations?

# Infinitely Many Solutions

Let us again recall what this case looked like when solving pairs of equations.

## Basic Exercise

Solve the following systems of equations with two variables both algebraically and graphically:

$$(a) \begin{cases} y = x + 1 \\ 2x - 2y + 2 = 0 \end{cases} \qquad (b) \begin{cases} y = -x + 2 \\ y = x \\ y = 2x - 1 \end{cases}$$

When the equations of a pair of equations with two variables are combined such that the variable  $y$  is eliminated, it is required that the lines corresponding to the equations are at the same height (where height is considered as  $y$ ). Thus, from the resulting equation with one variable, the points on the  $x$ -axis can be solved where this is the case.

In the previous task (a), proceeding in this way leads to the true equation  $0 = 0$ . This means that the lines are at the same height for every value of the variable  $x$ . Therefore, any point on the line  $y = x + 1$  can serve as a solution to the system of equations. These can be expressed, for example, in the form  $(x, y) = (x, x + 1)$ ,  $x \in \mathbb{R}$ , where the variable  $x$ , for which no condition was determined, is often referred to as a **free parameter**.

The previous solution can also be repeated by first eliminating  $x$ , leading to an equation that contains only the variable  $y$ . This further simplifies to the form  $0 = 0$ , which means the same as before: the system of equations is satisfied for any value of  $y$ , provided that  $x$  is in the form  $x = y - 1$ . Thus, all points of the form  $(x, y) = (y - 1, y)$ ,  $y \in \mathbb{R}$ , serve as solutions. In this solution,  $y$  is left as a free parameter.

The fact that both variables could be chosen as free parameters in the previous solutions is not an exception but rather arises from the fact that the line  $y = x + 1$  can also be written equivalently in the form  $x = y - 1$ . Reaching the equation  $0 = 0$  when solving the system of equations practically means that any variable for which no value is determined can be chosen as a free parameter, with the answer formulated using it.

### Example

Let us further express the solution of the previous system of equations in a clear form.

$$\begin{aligned} \begin{cases} y = x + 1 \\ 2x - 2y + 2 = 0 \end{cases} &\iff \begin{cases} y = x + 1 \\ 2x - 2(x + 1) + 2 = 0 \end{cases} \\ &\iff \begin{cases} y = x + 1 \\ 0 = 0 \end{cases}, \end{aligned}$$

thus we have  $y = x + 1$ , and the solution remains with a free parameter. Let us choose  $x$  as the free parameter, yielding the solution in the form

$$(x, y) = (x, x + 1), \quad x \in \mathbb{R}.$$

If the problem is solved using the Gauss algorithm, the equation  $0 = 0$  corresponds to a zero row, which usually indicates that the solution is parametric.

### Example

Let us repeat the previous example in matrix form while highlighting the reasoning steps.

When solving a  $2 \times 2$  system of equations, we want to determine the  $\begin{bmatrix} x & y \end{bmatrix}^T$  vectors that satisfy the system of equations.

First, we write the system of equations as an augmented matrix:

$$\begin{cases} y = x + 1 \\ 2x - 2y + 2 = 0 \end{cases} \iff \begin{cases} -x + y = 1 \\ 2x - 2y = -2 \end{cases}$$
$$\iff \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \iff \left[ \begin{array}{cc|c} -1 & 1 & 1 \\ 2 & -2 & -2 \end{array} \right],$$



### Example (continued)

We then write it in row-echelon form:

$$\left[ \begin{array}{cc|c} -1 & 1 & 1 \\ 2 & -2 & -2 \end{array} \right] \Longleftrightarrow \left[ \begin{array}{cc|c} -1 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right].$$

Thus, we have  $-x + y = 1$  and  $0 = 0$ , meaning the solution retains a free parameter. Let us choose  $x$  as the free parameter. At this point, we know that the solution is in the form

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ ? \end{bmatrix}, \quad x \in \mathbb{R},$$

so we still need to determine the expression for the variable  $y$ , which can contain the variable  $x$ . From the equation  $-x + y = 1$ , we have  $y = 1 + x$ , thus the solution is obtained as

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ x + 1 \end{bmatrix}, \quad x \in \mathbb{R}.$$

### Example (continued)

As a final step, let's write the solution as

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} x \\ x \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + x \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad x \in \mathbb{R},$$

from which it is more clearly recognized as the equation of a line.

The step from the previous solution

$$\left[ \begin{array}{cc|c} -1 & 1 & 1 \\ 2 & -2 & -2 \end{array} \right] \Longleftrightarrow \left[ \begin{array}{cc|c} -1 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right],$$

where the term corresponding to the variable  $x$  was eliminated from the second equation (and in this case, accidentally, all other terms as well), geometrically means determining the values of the variable  $y$  for which the lines are at the same height with respect to  $x$ . The answer is provided by the lower row, which in this case is a zero row, i.e., the equation  $0 = 0$ , meaning that the lines are at the same height for every value of the variable  $y$ . This of course also means that the lines are at the same height with respect to  $y$  for every value of the variable  $x$ .

In the previous example, the parametric solution was also expressed at the end in a form that identifies it as a line. Similarly, this can be done in the case of three or more variables.

The previous zero row indicates a parametric solution. However, this is not always the case. For example, in Problem (b), we obtain

$$\left[ \begin{array}{cc|c} 1 & 1 & 2 \\ -1 & 1 & 0 \\ -2 & 1 & -1 \end{array} \right] \Longleftrightarrow \left[ \begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 2 & 2 \\ 0 & 3 & 3 \end{array} \right] \Longleftrightarrow \left[ \begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right] \Longleftrightarrow \left[ \begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right],$$

which corresponds to the equations  $x = 1$ ,  $y = 1$ , and  $0 = 0$ . In this case, the first two rows give the intersection point of the first two equations, and since the third equation is also true, this point satisfies the third equation. Thus, the system of equations has a unique solution  $(x, y) = (1, 1)$ . This can happen when there are more equations than variables in the system. In other cases, a zero row indicates that the answer retains a free parameter unless there is a false equation in the system, in which case it has no solution.

On the other hand, the solution may still retain free parameters, meaning the equation may have a parametric solution even if no zero rows are encountered in the Gauss algorithm. If, in fact, the linear system has fewer equations than variables, the solution will always be parametric, provided a solution exists. Let us solve the following problem as an example using the Gauss algorithm.

### Example

$$\begin{cases} x + y + z = 0 \\ 2x - y + z = 1 \end{cases} \iff \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & -1 & 1 & 1 \end{array} \right] \iff \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -3 & -1 & 1 \end{array} \right] \\ \iff \begin{cases} x + y + z = 0 \\ -3y - z = 1 \end{cases}$$

Since the system of equations to be solved has three unknowns and two equations, one of the unknowns remains free. Thus, we need to choose a free parameter, solve the other two variables in relation to this, and write the solution points  $(x, y, z)$  concerning the free parameter.

### Example (continued)

Let us choose  $y$  as the free parameter, as it is easy to solve for  $x$  and  $z$  in relation to it. Then, from the lower equation, we have

$$-3y - z = 1 \iff z = -3y - 1,$$

after which from the upper equation, we get

$$x + y + z = 0 \iff x = -y - z = -y - (-3y - 1) = 2y + 1.$$

Thus, the solution is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2y + 1 \\ y \\ -3y - 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 2y \\ y \\ -3y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + y \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}, \quad y \in \mathbb{R}.$$

The number of free parameters is determined from the row echelon form by subtracting the number of non-zero rows from the number of variables. For example, in the previous system of equations, there are 3 variables and 2 non-zero rows, so the solution will have  $3 - 2 = 1$  free parameter. It is perhaps more natural to think of it this way: in the case of  $n$  variables,  $n$  equations are needed to obtain a unique solution, and each “missing” equation introduces one free parameter.

### Example

Solve the system of equations

$$\begin{cases} x + y + z = 0 \\ 2x - y + z = 1 \\ 4x + y + 3z = 1 \end{cases}.$$

## Solution

$$\begin{cases} x + y + z = 0 \\ 2x - y + z = 1 \\ 4x + y + 3z = 1 \end{cases} \iff \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & -1 & 1 & 1 \\ 4 & 1 & 3 & 1 \end{array} \right] \iff \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -3 & -1 & 1 \\ 0 & -3 & -1 & 1 \end{array} \right] \\ \iff \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -3 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

This system has 3 variables and its row echelon form has 2 non-zero rows. Thus, there will be  $3 - 2 = 1$  free parameter in the solutions.

The last form corresponds to the system of equations

$$\begin{cases} x + y + z = 0 \\ -3y - z = 1 \\ 0 = 0 \end{cases}$$

Thus, the same two equations hold for the variables as in the previous task, so the task can be solved to completion in the same way. Here, the zero row means that the solution retains a free parameter.

The step from the previous solution

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & -1 & 1 & 1 \\ 4 & 1 & 3 & 1 \end{array} \right] \Longleftrightarrow \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -3 & -1 & 1 \\ 0 & -3 & -1 & 1 \end{array} \right],$$

where the terms corresponding to the variable  $x$  were eliminated from the two latter equations, geometrically means determining the values of the variables  $y$  and  $z$  for which the first two planes, as well as the first and third planes, are at the same height with respect to  $x$ . The answer is given by the two bottom rows of the latter matrix.

The middle row, i.e.,  $-3y - z = 1$ , means that the first two planes are at the same height with respect to  $x$  along the line  $-3y - z = 1$  in the  $yz$ -plane. The third row tells the same thing for the first and third planes.



The step

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -3 & -1 & 1 \\ 0 & -3 & -1 & 1 \end{array} \right] \Longleftrightarrow \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -3 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

where the term corresponding to the variable  $y$  was eliminated from the last equation (and in this case, accidentally, all other terms as well), geometrically means determining the values of the variable  $z$  for which the lines  $-3y - z = 1$  and  $-3y - z = 1$  are at the same height with respect to  $y$ . The answer is given by the bottom row of the latter matrix, which in this case is a zero row, i.e., the equation  $0 = 0$ , meaning that the lines are at the same height with respect to  $y$  always, i.e., for every value of the variable  $z$ .

In the previous solutions, any variable could be chosen as the free parameter. This is not always the case, however. For example, in the case

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

we have  $z = 2$ , so the variable  $z$  cannot be chosen as a free parameter.

On the other hand, in the case

$$\left[ \begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

the free parameter must be chosen as  $x_4$  (You will notice why when you try to write the solution without this choice), but the second free parameter can be freely chosen from among the variables  $x_1$ ,  $x_2$ , and  $x_3$ .

There may also be several free parameters. If, in the case of three variables, two are left free, what is the geometric interpretation of the solution?

### Example

Solve the following systems of equations using Gauss-Jordan elimination:

$$(a) \begin{cases} -3x + 2y - 6z = 6 \\ 5x + 7y - 5z = 6 \\ x + 4y - 2z = 8 \end{cases}$$

$$(b) \begin{cases} 2x_1 - x_2 + x_3 = 1 \\ 3x_1 + 2x_2 - 4x_3 = 4 \\ -6x_1 + 3x_2 - 3x_3 = 2 \end{cases}$$

$$(c) \begin{cases} 2x_1 - x_2 + x_3 = 0 \\ 3x_1 + 2x_2 - 4x_3 = 0. \\ -6x_1 + 3x_2 - 3x_3 = 0 \end{cases}$$

## Solution

(a) First, we write the system of equations as an augmented matrix:

$$\begin{cases} -3x + 2y - 6z = 6 \\ 5x + 7y - 5z = 6 \\ x + 4y - 2z = 8 \end{cases} \iff \begin{bmatrix} -3 & 2 & -6 \\ 5 & 7 & -5 \\ 1 & 4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \\ 8 \end{bmatrix}$$
$$\iff \left[ \begin{array}{ccc|c} -3 & 2 & -6 & 6 \\ 5 & 7 & -5 & 6 \\ 1 & 4 & -2 & 8 \end{array} \right].$$

(Since there are three variables and three non-zero rows in the row echelon form, the solution will have  $3 - 3 = 0$  free parameters.) We will then solve the system of equations in this form using Gauss-Jordan elimination:

$$\left[ \begin{array}{ccc|c} -3 & 2 & -6 & 6 \\ 5 & 7 & -5 & 6 \\ 1 & 4 & -2 & 8 \end{array} \right] \iff \dots \iff \left[ \begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{array} \right].$$

## Solution (continue)

Now let's express the solution in terms of the variables

$$\left[ \begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{array} \right] \iff \begin{cases} x = -2 \\ y = 3 \\ z = 1 \end{cases}.$$

(b)

$$\begin{aligned} \left[ \begin{array}{ccc|c} 2 & -1 & 1 & 1 \\ 3 & 2 & -4 & 4 \\ -6 & 3 & -3 & 2 \end{array} \right] &\iff \left[ \begin{array}{ccc|c} 6 & -3 & 3 & 3 \\ 6 & 4 & -8 & 8 \\ -6 & 3 & -3 & 2 \end{array} \right] \\ &\iff \left[ \begin{array}{ccc|c} 6 & -3 & 3 & 3 \\ 0 & 7 & -11 & 5 \\ 0 & 0 & 0 & 5 \end{array} \right], \end{aligned}$$

and since the bottom row is false  $0 \cdot x + 0 \cdot y + 0 \cdot z = 5$ , the system of equations has no solution.

## Solution (continued)

(c)

$$\begin{aligned} \left[ \begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 3 & 2 & -4 & 0 \\ -6 & 3 & -3 & 0 \end{array} \right] &\iff \left[ \begin{array}{ccc|c} 6 & -3 & 3 & 0 \\ 0 & 7 & -11 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \\ &\iff \dots \iff \left[ \begin{array}{ccc|c} 1 & 0 & -\frac{2}{7} & 0 \\ 0 & 1 & -\frac{11}{7} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]. \end{aligned}$$

(Since there are three variables and two non-zero rows in the row echelon form, the solution will have  $3 - 2 = 1$  free parameter. Since none of the variables has a uniquely determined value, any of the variables  $x$ ,  $y$ , or  $z$  can be chosen as the free parameter.) Choose  $z$  as the free parameter, leading to the conditions

$$\begin{cases} x - \frac{2}{7}z = 0 \\ y - \frac{11}{7}z = 0 \end{cases} \iff \begin{cases} x = \frac{2}{7}z \\ y = \frac{11}{7}z \end{cases}, \quad z \in \mathbb{R}.$$

### Solution (continued)

The solution is obtained as

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{2}{7}z \\ \frac{11}{7}z \\ z \end{bmatrix} = z \begin{bmatrix} \frac{2}{7} \\ \frac{11}{7} \\ 1 \end{bmatrix}, \quad z \in \mathbb{R},$$

which is a line passing through the origin.

### Basic Exercise

Solve the following  $3 \times 3$  systems of equations and give the geometric interpretations of the solutions:

$$(a) \begin{cases} z = x + 1 \\ x + z = 3 \\ 3x - z - 1 = 0 \end{cases}$$

$$(b) \begin{cases} z = x + 1 \\ x + z = 3 \\ 3x - z - 5 = 0 \end{cases}$$

$$(c) \begin{cases} x + 2y + 3z = 4 \\ 2(y - x - 2) = -3(z + x) \\ -2y - 9 + 5z = 3 - 3x - 4z - 8y. \end{cases}$$

Multivariable solutions generalize similarly to situations with multiple variables: if the solution of a system of equations has  $n$  free parameters, the solution is in the form

$$\mathbf{x} = \tilde{\mathbf{b}} + t_1 \mathbf{v}_1 + t_2 \mathbf{v}_2 + \cdots + t_n \mathbf{v}_n, \quad t_1, \dots, t_n \in \mathbb{R},$$

where the coefficients  $t_i$  are free parameters.



### Example

The equation

$$\begin{bmatrix} 1 & 0 & -4 \\ -2 & 0 & 3 \\ -1 & 2 & 1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ a \end{bmatrix} \quad (1)$$

describes a certain process where  $\mathbf{x}$  represents the quantities of raw materials, and the vector on the right-hand side expresses the desired quantities of final products. For which values of the parameter  $a$  does the equation

- (a) have one solution,
- (b) have no solutions at all,
- (c) have infinitely many solutions?

### Solution

Let's try to solve the system of equations using Gaussian elimination and see how the solution depends on the constant  $a$ :

$$\begin{bmatrix} 1 & 0 & -4 \\ -2 & 0 & 3 \\ -1 & 2 & 1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ a \end{bmatrix}$$

## Solution (continued)

$$\begin{aligned}
 & \left[ \begin{array}{ccc|c} 1 & 0 & -4 & 1 \\ -2 & 0 & 3 & 1 \\ -1 & 2 & 1 & a \end{array} \right] \begin{array}{l} +2R_1 \\ +R_1 \end{array} \\
 \sim & \left[ \begin{array}{ccc|c} 1 & 0 & -4 & 1 \\ 0 & 0 & -5 & 3 \\ 0 & 2 & -3 & a+1 \end{array} \right] R_2 \leftrightarrow R_3 \\
 \sim & \left[ \begin{array}{ccc|c} 1 & 0 & -4 & 1 \\ 0 & 2 & -3 & a+1 \\ 0 & 0 & -5 & 3 \end{array} \right] \cdot \left(-\frac{1}{5}\right) \\
 \sim & \left[ \begin{array}{ccc|c} 1 & 0 & -4 & 1 \\ 0 & 2 & -3 & a+1 \\ 0 & 0 & 1 & -\frac{3}{5} \end{array} \right] \begin{array}{l} +4R_3 \\ +3R_3 \end{array} \\
 \sim & \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 1 - \frac{12}{5} \\ 0 & 2 & 0 & a+1 - \frac{9}{5} \\ 0 & 0 & 1 & -\frac{3}{5} \end{array} \right] \cdot \left(\frac{1}{2}\right) \\
 \sim & \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 1 - \frac{12}{5} \\ 0 & 1 & 0 & \frac{1}{2}a - \frac{4}{10} \\ 0 & 0 & 1 & -\frac{3}{5} \end{array} \right]
 \end{aligned}$$

### Solution (continued)

$$\begin{cases} x_1 = -\frac{7}{5} \\ x_2 = \frac{1}{2}a - \frac{2}{5} \\ x_1 = -\frac{3}{5} \end{cases}$$

We observe that for any value of the constant  $a$ , there is exactly one solution.

### Example

Let's investigate the system of equations  $A\mathbf{x} = \mathbf{b}$ . The augmented matrix representing the system  $\tilde{A} = [A \ \mathbf{b}]$  can be transformed into row echelon form using Gaussian elimination. Let's denote the resulting echelon matrix with the symbol  $B$ . We will investigate five different systems of equations:

$$\begin{aligned} \text{(i)} \quad B &= \begin{bmatrix} 2 & 4 & 0 \\ 0 & 2 & 1 \end{bmatrix} & \text{(ii)} \quad B &= \begin{bmatrix} 2 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} & \text{(iii)} \quad B &= \begin{bmatrix} 2 & 4 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ \text{(iv)} \quad B &= \begin{bmatrix} 2 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} & \text{(v)} \quad B &= \begin{bmatrix} 2 & 4 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

For each case, answer the following questions:

- (a) Is any of the systems examined in the previous part homogeneous? If not, how would the matrix  $B$  change if the original system of equations were homogeneous (but  $A$  did not change)?
- (b) Solve the system of equations represented by the matrix  $B$ . If possible, give the solution in the form  $\mathbf{x} = \tilde{\mathbf{b}} + c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$ , where  $c_i \in \mathbb{R}$  (that is, find the vectors  $\tilde{\mathbf{b}}$  and  $\mathbf{v}_i$ ).
- (c) Interpret the previous solutions geometrically for matrices (i)–(iii).
- (d) Solve the corresponding homogeneous systems of equations. Interpret the solutions geometrically for cases (i)–(iii).

## Solution

(a) No, because the last column of matrix B should be the zero vector.

(b) (i)

$$\left[ \begin{array}{cc|c} 2 & 4 & 0 \\ 0 & 2 & 1 \end{array} \right] \Longleftrightarrow \begin{cases} 2x_1 + 4x_2 = 0 \\ 2x_2 = 1 \end{cases} \Longleftrightarrow \dots \Longleftrightarrow \begin{cases} x_1 = -1 \\ x_2 = \frac{1}{2} \end{cases},$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \underbrace{\begin{bmatrix} -1 \\ \frac{1}{2} \end{bmatrix}}_{\tilde{\mathbf{b}}},$$

(ii)

$$\left[ \begin{array}{cc|c} 2 & 4 & 0 \\ 0 & 0 & 1 \end{array} \right] \Longleftrightarrow \begin{cases} 2x_1 + 4x_2 = 0 \\ 0 = 1 \end{cases}$$

Since the lower equation is false, the system of equations has no solution.

### Solution (continued)

(iii) From the augmented matrix

$$\left[ \begin{array}{ccc|c} 2 & 4 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

only one condition  $2x_1 + 4x_2 + 0 \cdot x_3 = 2$  is obtained. There are three variables and only one non-zero row in the row echelon form, so the solution has two free parameters. Since there is no  $x_3$  in the equations of row echelon form, it must be chosen as a free parameter. The other one can be either  $x_1$  or  $x_2$ . Let's choose  $x_2$  and solve the previous condition with respect to it:  $x_1 = 1 - 2x_2$ . The solution then takes the form

$$\begin{aligned} \mathbf{x} &= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 - 2x_2 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -2x_2 \\ 1 \cdot x_2 \\ 0 \cdot x_2 \end{bmatrix} + \begin{bmatrix} 0 \cdot x_3 \\ 0 \cdot x_3 \\ 1 \cdot x_3 \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{\tilde{\mathbf{b}}} + x_2 \underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{v}_1} + x_3 \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{\mathbf{v}_2}, \quad x_2, x_3 \in \mathbb{R}. \end{aligned}$$

### Solution (continued)

(iv) From the augmented matrix

$$\left[ \begin{array}{cccc|c} 2 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

the conditions are obtained

$$\begin{cases} 2x_1 + 4x_2 + x_4 = 0 \\ x_3 + x_4 = 1 \end{cases},$$

so there are two free parameters left in the solution. Let's choose these to be  $x_2$  and  $x_4$  and solve the previous conditions with respect to them:

$$\begin{cases} x_1 = -2x_2 - \frac{1}{2}x_4 \\ x_3 = 1 - x_4 \end{cases}.$$

The solution thus takes the form

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2x_2 - \frac{1}{2}x_4 \\ x_2 \\ 1 - x_4 \\ x_4 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}}_{\tilde{\mathbf{b}}} + x_2 \underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}}_{\mathbf{v}_1} + x_4 \underbrace{\begin{bmatrix} -\frac{1}{2} \\ 0 \\ -1 \\ 1 \end{bmatrix}}_{\mathbf{v}_2}, \quad x_2, x_4 \in \mathbb{R}.$$

Solution (continued)  
(v)

$$\begin{bmatrix} 2 & 4 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

$$\begin{cases} 2x_1 + 4x_2 + x_4 = 1 \\ x_3 + x_4 = 0 \\ 2x_4 = 4 \end{cases} \iff \begin{cases} 2x_1 + 4x_2 + 2 = 1 \\ x_3 + 2 = 0 \\ x_4 = 2 \end{cases}$$

$$\iff \begin{cases} x_1 = -\frac{1}{2} - 2x_2 \\ x_3 = -2 \\ x_4 = 2 \end{cases},$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} - 2x_2 \\ x_2 \\ -2 \\ 2 \end{bmatrix} = \underbrace{\begin{bmatrix} -\frac{1}{2} \\ 0 \\ -2 \\ 2 \end{bmatrix}}_{\tilde{\mathbf{b}}} + x_2 \underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}}_{\mathbf{v}_1}$$



## Solution (continued)

- (c) The solution to part (i) is a point. The solution to part (iii) is a plane (which does not pass through the origin).
- (d) In corresponding homogeneous systems of equations, the last column of the B matrix contains only zeros. This can lead to two types of changes in the previous solutions: (1) Each system of equations now has solutions. (2) Previous solutions change only in that the vector  $\tilde{\mathbf{b}}$  appearing in the answer becomes the zero vector. (How does this change the geometric interpretation of the solution in part (iii)?)

(i)

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

(ii)

$$\begin{bmatrix} 2 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies 2x_1 + 4x_2 = 0 \iff x_1 = -2x_2$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad x_2 \in \mathbb{R}$$

## Solution (continued)

(iii)

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

(iv)

$$\mathbf{x} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -\frac{1}{2} \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

(v)

$$\mathbf{x} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

## MATLAB Example

### Systems of equations

$$(a) \begin{cases} -3x + 2y - 6z = 6 \\ 5x + 7y - 5z = 6 \\ x + 4y - 2z = 8 \end{cases}$$

$$(b) \begin{cases} 2x_1 - x_2 + x_3 = 1 \\ 3x_1 + 2x_2 - 4x_3 = 4 \\ -6x_1 + 3x_2 - 3x_3 = 2 \end{cases}$$

$$(c) \begin{cases} 2x_1 - x_2 + x_3 = 0 \\ 3x_1 + 2x_2 - 4x_3 = 0 \\ -6x_1 + 3x_2 - 3x_3 = 0 \end{cases}$$

are brought into row echelon form in MATLAB (from which their solutions can be obtained easily) using the following commands:

## MATLAB Example (continued)

```
A1 = [-3, 2, -6; 5, 7, -5; 1, 4, -2];  
b1 = [6; 6; 8];  
C1 = [A1 b1];  
x1 = rref(C1) % Reduced Row Echelon Form
```

```
A2 = [2, -1, 1; 3, 2, -4; -6, 3, -3];  
b2 = [1; 4; 2];  
C2 = [A2 b2];  
x2 = rref(C2)
```

```
b3 = [0; 0; 0];  
C3 = [A2 b3];  
x3 = rref(C3)
```

# Linear Systems of Equations and Coordinate Transformations

In this chapter, we examine how solutions of systems of equations can be understood through coordinate transformations.

**Unique Solution.** Recall that the equation

$$\begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

can also be written in the form

$$0 \begin{bmatrix} -1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The latter equation can be interpreted as the matrix  $\begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}$  transforming the point  $(0, 2)$  into a new coordinate system whose basis vectors are  $\begin{bmatrix} -1 & 0 \end{bmatrix}^T$  and  $\begin{bmatrix} 1 & 1 \end{bmatrix}^T$ , and in this coordinate system, it corresponds to the standard coordinate point  $(2, 2)$ .

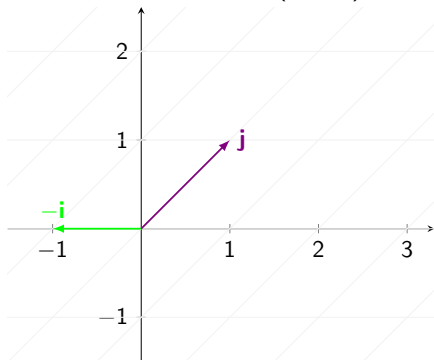
In a similar way, the system of equations

$$\begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

can be written in the form

$$x \begin{bmatrix} -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

The latter equation can be interpreted as a question of what point  $(x, y)$  in the coordinate system spanned by the vectors  $\begin{bmatrix} -1 & 0 \end{bmatrix}^T$  and  $\begin{bmatrix} 1 & 1 \end{bmatrix}^T$  corresponds to the standard coordinate point  $(3, 1)$ . Conclude from the accompanying image that such a point is  $(-2, 1)$ .



More generally, the system of equations

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix},$$

where  $\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} = \mathbf{A}$ ,  $\begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}^T = \mathbf{x}$ , and  $\begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix}^T = \mathbf{b}$ , can be written in the form

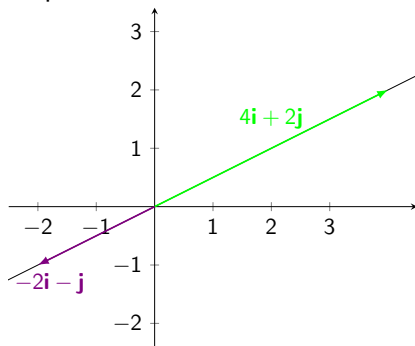
$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix},$$

and the latter equation can be interpreted as a question of what point  $(x_1, x_2, \dots, x_n)$  in the coordinate system spanned by the vectors  $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$  corresponds to the standard coordinate point  $(b_1, b_2, \dots, b_n)$ . Next, let's consider what it means from this perspective if a system of equations has no solution or if the solution is ambiguous.

**No Solution.** The system of equations

$$\begin{bmatrix} 4 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

can be interpreted as a question of what point  $(x, y)$  in the coordinate system spanned by the vectors  $\begin{bmatrix} 4 & 2 \end{bmatrix}^T$  and  $\begin{bmatrix} -2 & -1 \end{bmatrix}^T$  corresponds to the standard coordinate point  $(1, 2)$ . The accompanying image shows that the space spanned by these basis vectors is not the entire  $xy$ -plane but rather a line passing through the origin. Since the point  $(1, 2)$  does not lie on this line, the system of equations has no solution. Confirm this by solving the system of equations.





That the basis vectors span a line instead of a plane can be seen concretely from their linear combination as follows: Let  $x$  and  $y$  be arbitrary real numbers. Then all points in the coordinate system spanned by the vectors  $\begin{bmatrix} 4 & 2 \end{bmatrix}^T$  and  $\begin{bmatrix} -2 & -1 \end{bmatrix}^T$  can be obtained by the linear combination

$$x \begin{bmatrix} 4 \\ 2 \end{bmatrix} + y \begin{bmatrix} -2 \\ -1 \end{bmatrix} = 2x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + (-1)y \begin{bmatrix} 2 \\ 1 \end{bmatrix} = (2x - y) \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad x, y \in \mathbb{R}.$$

Since  $x$  and  $y$  were arbitrary, the expression  $2x - y$  can take any value. Thus, we can use a simpler notation  $t$ , where  $t \in \mathbb{R}$ . Therefore, we have

$$x \begin{bmatrix} 4 \\ 2 \end{bmatrix} + y \begin{bmatrix} -2 \\ -1 \end{bmatrix} = t \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R},$$

from which it can be seen that the linear combination is a line passing through the origin rather than a plane.

This can be summarized as follows: the system of equations  $A\mathbf{x} = \mathbf{b}$  has no solution if the vector  $\mathbf{b}$  does not belong to the space spanned by the column vectors of the matrix  $A$ . This is quite evident when viewed through functions. The expression  $A_{m \times n}\mathbf{x}$  can be interpreted as a function

$$\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m, \mathbf{f}(\mathbf{x}) = A\mathbf{x}.$$

Thus, the system of equations  $A\mathbf{x} = \mathbf{b}$  can be written in the form  $\mathbf{f}(\mathbf{x}) = \mathbf{b}$ . This system of equations has no solution if  $\mathbf{b}$  does not belong to the range of the function  $\mathbf{f}$ . Compare this to the fact that the equation  $\cos(x) = 2$  cannot have a solution. It can also be noted that the system of equations  $A\mathbf{x} = \mathbf{b}$  may not have a solution if  $\mathbf{f}$  is not surjective. In the previous example

$$\mathbf{f}: \mathbb{R}^2 \rightarrow \mathbb{R}^2, \mathbf{f}(\mathbf{x}) = \begin{bmatrix} 4 & -2 \\ 2 & -1 \end{bmatrix} \mathbf{x},$$

was not surjective because its range is  $\mathcal{R}(\mathbf{f}) = \{ (x, y) \in \mathbb{R}^2 \mid y = x/2 \}$  and not  $\mathbb{R}^2$ .

Let's take a couple of examples in the case of three variables.

## Example

The system of equations

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

has no solution because there are three parallel planes at different heights. The fact that a solution is likely not to exist could also have been observed from the fact that the column vectors of the coefficient matrix span a line instead of the entire space  $\mathbb{R}^3$ :

$$x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} + z \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 2y \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 3z \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = (x+2y+3z) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

for  $t \in \mathbb{R}$ . The system of equations has no solution because the vector on the right side does not lie on this line.

### Example

In the case of

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

the planes are not parallel, and the column vectors of the coefficient matrix are not multiples of each other, so one might think that the system of equations likely has a solution. However, based on the linear combination of the column vectors of the coefficient matrices, it can be observed that they span a plane instead of the entire space  $\mathbb{R}^3$ :

$$\begin{aligned} x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + z \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} &= x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + z \left( 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right) \\ &= (x + 2z) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + (y + z) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = s \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad s, t \in \mathbb{R}. \end{aligned}$$

The vectors lie on the same plane, and thus their linear combination cannot escape that plane. Hence, the system of equations has no solution if the vector on the right side does not lie on that plane. In this case, it doesn't, but for example, the vector  $\begin{bmatrix} 2 & 3 & 4 \end{bmatrix}^T$  could happen to lie on it.

In the examples of this chapter, there have been vectors that are superfluous from the perspective of linear combinations. Particularly in the last example, there was such a vector, even though none of the vectors were multiples of each other. Although this will be discussed in more detail later, it can be noted at this stage that such a set of vectors is called **linearly dependent**, while a set of vectors that does not have superfluous vectors in terms of linear combinations is called **linearly independent**. The linear dependence of vectors means that the set of vectors can be expressed as a linear combination of a set of other vectors, as was the case with  $\begin{bmatrix} 3 & 4 & 5 \end{bmatrix}^T$  in the last example.

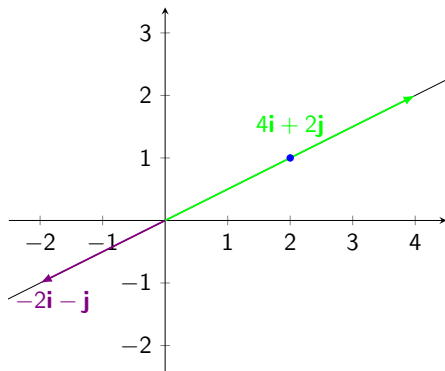
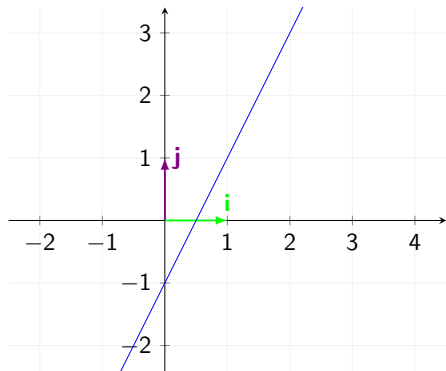
**Infinitely Many Solutions.** What if, in the examples of the previous chapter, the point on the right side of the system of equations had actually landed in the space spanned by the new basis vectors, that is, the column vectors of the coefficient matrix? This is the case, for example, with the system of equations

$$\begin{bmatrix} 4 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

(Inferring from the image.) In this case, the solution becomes multivalued. Namely, the fact that the coordinate transformation maps the  $xy$ -plane to a line is only possible when multiple points map to the same point. For instance, the solution to the previous system of equations is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} + x \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad x \in \mathbb{R},$$

which means that all points on the original coordinate system's line  $y = 2x - 1$  map to the point  $(2, 1)$ . This is illustrated in the accompanying images.



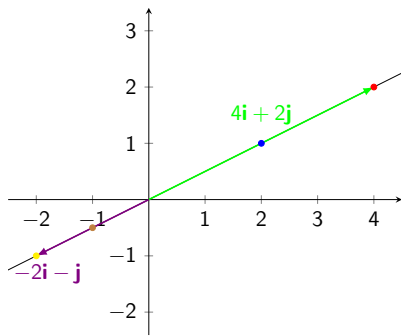
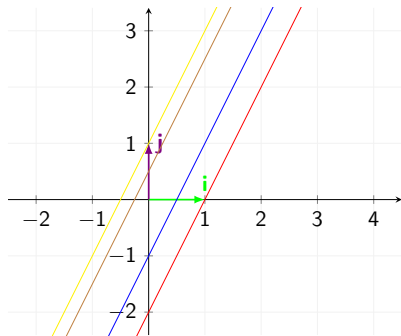
More generally, it holds that all points on the line

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ a \end{bmatrix} + x \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad x \in \mathbb{R},$$

map to the point  $(-2a, -a)$ , which is seen from the calculation

$$\begin{bmatrix} 4 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ 2x + a \end{bmatrix} = \begin{bmatrix} -2a \\ -a \end{bmatrix}.$$

This is illustrated in the accompanying images.



Together, this leads to the  $xy$ -plane being mapped to the line  $y = 2x - 1$ .



This can be summarized as follows: the system of equations  $A\mathbf{x} = \mathbf{b}$  has a multivalued solution if the vector  $\mathbf{b}$  belongs to the space spanned by the column vectors of the matrix  $A$  and this space is narrower than the one to which the vector  $\mathbf{x}$  belongs.

## Inverse Matrix

# Inverse Matrix

The inverse of an  $n \times n$  matrix  $A$  is an  $n \times n$  matrix  $B$  such that

$$AB = BA = I.$$

The inverse matrix is denoted by  $A^{-1}$ .

Thus, based on the previous definition, it follows that  $AA^{-1} = A^{-1}A = I$ . (In the definition, the notation  $B$  is used instead of  $A^{-1}$  because at this stage, it is not yet known what the notation  $A^{-1}$  would imply. It refers to the matrix that satisfies the equations mentioned in the definition.)

Notice the analogy between the inverse matrix and the reciprocal of a real number  $a \neq 0$ . However, there are two shortcomings in this analogy. First, there are other cases besides  $A = O$  where an inverse matrix does not exist. (These cases can be framed under the condition  $\det(A) \neq 0$ , which is discussed in connection with determinants.) Second, multiplying by the reciprocal corresponds to division, that is,

$$a^{-1}b = ba^{-1} = \frac{b}{a},$$

but this cannot be written for matrices since generally, it does not hold that  $A^{-1}B = BA^{-1}$ . Therefore, we **CANNOT** write

$$A^{-1}B = \frac{B}{A},$$

because it is not clear from which side the matrix  $A^{-1}$  is multiplied in the division notation. **There is no division for matrices.**

## Basic Exercise

Which of the following matrices is the inverse matrix of

$$\begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}?$$

(a)  $\begin{bmatrix} -1 & 1 \\ -4 & 3 \end{bmatrix}$

(b)  $\begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix}$

(c)  $\begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}?$

## Exercise

Let  $A = \begin{bmatrix} 3 & 1 \\ -2 & 4 \end{bmatrix}$ . Formulate a system of linear equations that can be solved to obtain the inverse matrix  $A^{-1}$ .

The inverse matrix can be determined, for example, using Gauss-Jordan elimination. This is done by forming the augmented matrix  $\tilde{A} = [A \quad I]$  and transforming it into the form  $[I \quad K]$ , where  $K = A^{-1}$ .

### Example

Determine the inverse matrix of

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 5 & 5 & 0 \end{bmatrix}.$$

### Solution

As previously mentioned, in the Gaussian algorithm, it can be convenient to multiply fractional rows by integers, which simplifies the addition of elements. This is particularly true when determining the inverse matrix, where more calculations need to be performed than when solving a regular system of equations. In the following solution, we will proceed this way.

# Solution (continued)

$$\begin{aligned}
 & \left[ \begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 5 & 5 & 0 & 0 & 0 & 1 \end{array} \right] \quad -5R_1 \\
 \sim & \left[ \begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 10 & -5 & -5 & 0 & 1 \end{array} \right] \quad R_2 \leftrightarrow R_3 \\
 \sim & \left[ \begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 10 & -5 & -5 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array} \right] \quad \begin{array}{l} -R_3 \\ +5R_3 \end{array} \\
 \sim & \left[ \begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & -1 & 0 \\ 0 & 10 & 0 & -5 & 5 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array} \right] \quad \cdot 10 \\
 \sim & \left[ \begin{array}{ccc|ccc} 10 & -10 & 0 & 10 & -10 & 0 \\ 0 & 10 & 0 & -5 & 5 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array} \right] \quad +R_2 \\
 \sim & \left[ \begin{array}{ccc|ccc} 10 & 0 & 0 & 5 & -5 & 1 \\ 0 & 10 & 0 & -5 & 5 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array} \right] \quad \begin{array}{l} \cdot \frac{1}{10} \\ \cdot \frac{1}{10} \end{array} \\
 \sim & \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{10} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{10} \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array} \right]
 \end{aligned}$$

## Solution (continued)

Hence

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{10} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{10} \\ 0 & 1 & 0 \end{bmatrix}$$

If the solution had divided the bottom row by 5 right away, it would have led to dealing with error-prone fractions already in the second step.

## Basic Exercise

Determine the inverse matrices of the following matrices:

(a)  $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

(b)  $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 0 \\ 0 & 1 & 2 \end{bmatrix}$

(c)  $\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ .



An inhomogeneous system of equations may not have a unique solution, which means that Gaussian elimination does not always reach the form  $[I \ K]$  when solving for the inverse matrix. This indicates that not all matrices have an inverse. If a matrix  $A$  has an inverse, the matrix is called **regular**, **non-singular**, or **invertible**; otherwise, it is referred to as **irregular** or **singular**. The existence of an inverse matrix can be simply stated using the determinant: a matrix  $A$  has an inverse if and only if  $\det(A) \neq 0$ . This will be discussed further in the chapter on determinants. At the same time, a method will be presented for determining the inverse matrix using determinants. However, the previously described method is more convenient.

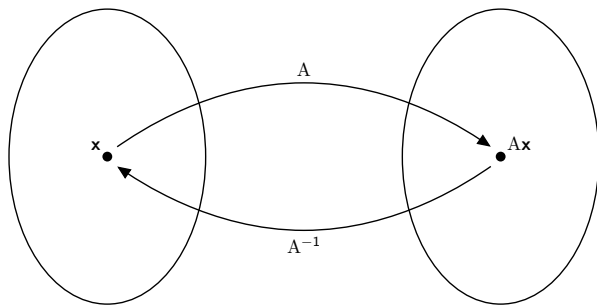


Figure: Inverse Matrix

What is the inverse matrix used for? From the definition of the inverse matrix, it follows that if the inverse matrix  $A^{-1}$  exists, then the system of equations  $A\mathbf{x} = \mathbf{b}$  can be solved by multiplying it by the inverse matrix from the left:

$$\begin{aligned}A\mathbf{x} &= \mathbf{b} && | \cdot A^{-1} \text{ from the left} \\A^{-1}A\mathbf{x} &= A^{-1}\mathbf{b} \\I\mathbf{x} &= A^{-1}\mathbf{b} \\\mathbf{x} &= A^{-1}\mathbf{b}.\end{aligned}$$

Note that multiplication by the inverse matrix must be done from the left, as multiplying from the right leads to the expression  $A\mathbf{x}A^{-1}$ , which cannot be simplified. Compare the previous calculation to the solution of a similar equation in the case of real numbers:

$$\begin{aligned}ax &= b \\a^{-1}ax &= a^{-1}b \\1 \cdot x &= a^{-1}b \\x &= a^{-1}b.\end{aligned}$$

Thus, if the inverse matrix of the coefficient matrix is available, the system of equations can be solved directly as a multiplication of a matrix and a vector. Furthermore, each value of the vector  $\mathbf{x}$  can be calculated independently of each other, allowing for efficient computation, for example, by using parallel computing.

A few properties related to the inverse matrix:

- The inverse matrix is unique.
- A matrix  $A$  has an inverse matrix if and only if  $\det(A) \neq 0$ .
- The inverse of the inverse matrix:  $(A^{-1})^{-1} = A$ .
- The inverse of the transpose:  $(A^T)^{-1} = (A^{-1})^T$ .
- The inverse of a scalar multiple:  $(kA)^{-1} = k^{-1}A^{-1}$ .
- The inverse of a product:  $(AB)^{-1} = B^{-1}A^{-1}$ .
- If  $A$  is a singular  $n \times n$  matrix and  $B$  is an  $n \times n$  matrix, then both  $AB$  and  $BA$  are also singular.

### Example

Solve the system of equations  $A\mathbf{x} = \mathbf{b}$  using the inverse matrix, where

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 5 & 5 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}.$$

### Solution

We have already computed the inverse matrix of  $A$  in an earlier example. Since the inverse matrix exists, we can write

$$A\mathbf{x} = \mathbf{b}$$

$$A^{-1}A\mathbf{x} = A^{-1}\mathbf{b}$$

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

Substituting the previously computed  $A^{-1}$  and  $\mathbf{b}$  gives us

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{10} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{10} \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{9}{10} \\ -\frac{11}{10} \\ 0 \end{bmatrix}.$$

There are some essential differences in solving matrix equations compared to equations with real numbers:

- You cannot divide with matrices; instead, you must multiply by the inverse matrix. It is important to know that the inverse matrix exists.
- The zero product property does not hold (the condition  $A = O$  or  $B = O \implies AB = O$  still holds).
- It matters from which side the matrix is multiplied in the equation or from which side a common factor is taken.
- When factoring out a common factor, you may need to add the identity matrix to keep the calculations defined.

### Example

Assume that  $A$  and  $B$  are non-singular  $3 \times 3$  matrices and that  $\mathbf{x}$ ,  $\mathbf{y}$ , and  $\mathbf{b}$  are  $3 \times 1$  matrices (column vectors).

- (a) Solve for  $\mathbf{x}$ , given that  $A\mathbf{b} + B(\mathbf{x} - \mathbf{y}) = \mathbf{b}$ .
- (b) Under what conditions does the equation  $A\mathbf{b} + B(\mathbf{x} - \mathbf{y}) = \mathbf{x}$  have a unique solution with respect to the vector  $\mathbf{x}$ ?

### Solution

(a)

$$A\mathbf{b} + B(\mathbf{x} - \mathbf{y}) = \mathbf{b}$$

$$B(\mathbf{x} - \mathbf{y}) = \mathbf{b} - A\mathbf{b} \quad | \cdot B^{-1} \text{ from the left}$$

$$B^{-1}B(\mathbf{x} - \mathbf{y}) = B^{-1}(\mathbf{b} - A\mathbf{b})$$

$$I(\mathbf{x} - \mathbf{y}) = B^{-1}(\mathbf{b} - A\mathbf{b})$$

$$\mathbf{x} - \mathbf{y} = B^{-1}(\mathbf{b} - A\mathbf{b})$$

$$\mathbf{x} = \mathbf{y} + B^{-1}(\mathbf{b} - A\mathbf{b})$$

## Solution (continued)

(b)

$$A\mathbf{b} + B(\mathbf{x} - \mathbf{y}) = \mathbf{x}$$

$$B(\mathbf{x} - \mathbf{y}) - \mathbf{x} = -A\mathbf{b}$$

$$B\mathbf{x} - \mathbf{x} = -A\mathbf{b} + B\mathbf{y}$$

$$B\mathbf{x} - I\mathbf{x} = -A\mathbf{b} + B\mathbf{y}$$

$$\underbrace{(B - I)}_C \mathbf{x} = -A\mathbf{b} + B\mathbf{y}$$

$$C\mathbf{x} = -A\mathbf{b} + B\mathbf{y}$$

The equation has a unique solution if and only if  $C^{-1}$  exists, in which case the solution is given by

$$\underbrace{C^{-1}C}_I \mathbf{x} = C^{-1}(-A\mathbf{b} + B\mathbf{y})$$

$$\mathbf{x} = C^{-1}(-A\mathbf{b} + B\mathbf{y}).$$



If there are problems solving the matrix equation, it is advisable to first consider how solving the corresponding ordinary equation would occur.

### Basic Exercise

(a) Solve the equation

$$\frac{ax + b}{cx + d} = y$$

for the variable  $x$  and state under what assumptions this solution exists.

(b) Solve the equation

$$(AX + B)(CX + D)^{-1} = Y$$

for the variable  $X$  and state under what assumptions this solution exists.

## Basic Exercise

When does the equation  $AA\mathbf{x} + \mathbf{y} + AB\mathbf{x} = \mathbf{0}$  have a unique solution with respect to the vector  $\mathbf{x}$ ? What is the solution of the equation in this case? Is the solution of the system unique in the case

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} ?$$

## Exercise

A typical application of the inverse matrix is the method of least squares, where a model (function) is fitted to data points  $(x_i, y_i)$ ,  $i = 1, \dots, n$ , in the form

$$y = a_1 f_1(x) + a_2 f_2(x) + \cdots + a_m f_m(x), \quad (2)$$

where  $m \leq n$ . If this problem has a unique solution, i.e., if unique values for the coefficients  $a_1, \dots, a_m$  are obtained, they are found by solving the system  $A^T A \mathbf{a} = A^T \mathbf{b}$ , where  $A \mathbf{a} = \mathbf{b}$  represents the system obtained by considering equation (2) for each data point  $(x_i, y_i)$ ,  $i = 1, \dots, n$ .

- (a) What are the dimensions of the matrices  $A$  and  $A^T A$ ?
- (b) Form the matrix  $A$  and the vector  $\mathbf{b}$  when fitting a linear model  $y = a_1 x$  to the data points  $(1, 1)$ ,  $(2, 3)$ , and  $(3, 2)$ .
- (c) Form the matrix  $A$  and the vector  $\mathbf{b}$  when fitting a linear model  $y = a_1 x + a_2$ .

## Matrix Inverse in the Special $2 \times 2$ Case

The  $2 \times 2$  matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad (3)$$

has a matrix inverse if and only if  $ad - bc \neq 0$ , in which case it is given by

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}. \quad (4)$$

### Exercise

Show that (4) is the matrix inverse of (3) when  $ad - bc \neq 0$ .

For larger matrices, it is more practical to use Gauss–Jordan elimination to compute the matrix inverse by hand.

## MATLAB Example

The system of equations

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 5 & 5 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

is solved in MATLAB using the inverse matrix as follows:

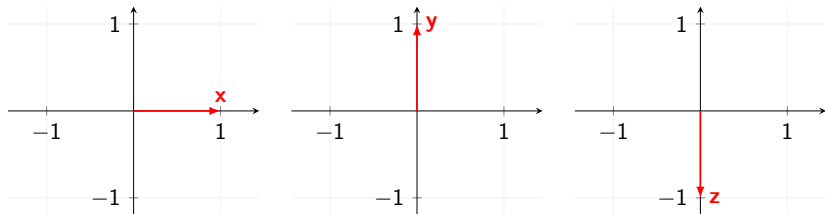
```
A = [1, -1, 1; 0, 0, 1; 5, 5, 0];  
b = [2; 0; 1];  
x = inv(A)*b
```

# Matrix Multiplication and Coordinate Transformations

We will consider the coordinate transformation corresponding to matrix multiplication, after which the treatment of the coordinate transformation corresponding to the inverse matrix will become straightforward. Let

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The first matrix reflects a point (vector) with respect to the line  $y = x$ , and the second one reflects it with respect to the  $x$ -axis. The mapping  $B(A\mathbf{x}) = B\mathbf{y} = \mathbf{z}$  thus geometrically means reflecting the vector  $\mathbf{x}$  with respect to the line  $y = x$  and then reflecting the resulting vector  $\mathbf{y}$  with respect to the  $x$ -axis. This has been illustrated in the accompanying images for the case of the vector  $\mathbf{x} = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ .



We will check by calculating that the result works as concluded:

$$B(A\mathbf{x}) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \left( \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Since the matrix multiplication rules state that  $B(A\mathbf{x}) = (BA)\mathbf{x}$ , the product matrix  $BA$  performs the same coordinate transformation as applying  $A$  and then  $B$ . We can confirm this by calculating in the case of the previous example:

$$(BA)\mathbf{x} = \left( \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right) \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

However, the order in which coordinate transformations are applied matters, as observed in the following exercise.

### Basic Exercise

Deduce, based on the previously discussed matrices  $A$  and  $B$  and the vector  $\mathbf{x}$ , what  $AB\mathbf{x}$  is (a) based on coordinate transformations (b) by calculation.

This geometrically illustrates why the property  $AB = BA$  does not generally hold for matrix multiplication.

# Inverse Matrix and Coordinate Transformations

If the matrix  $A$  corresponds to a certain coordinate transformation, then its inverse matrix, provided it exists, corresponds to the reverse coordinate transformation. This can be concluded from the fact that  $A^{-1}(A\mathbf{x}) = \mathbf{x}$ , meaning that  $A^{-1}$  cancels out the effect of matrix  $A$ .

## Exercise

Deduce based on coordinate transformations what the inverse matrix of  $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$  is and then confirm your reasoning by calculation.

Coordinate transformations provide a geometric explanation for why an inverse matrix does not always exist: this occurs in cases where the coordinate transformation corresponding to the inverse matrix would need to map each point to multiple points. For example, the matrix  $\begin{bmatrix} 4 & -2 \\ 2 & -1 \end{bmatrix}$  discussed in Chapter 4.4.5.3 maps every point on the line  $y = 2x - 1$  to the point  $(2, 1)$ , so its inverse matrix would need to map this point back to a line, which is not possible. This is most clearly understood through the lens of functions, which will be examined in the next chapter.



## Determinants

Determinants have already been introduced in the sections dealing with the cross product and inverse matrices. In this chapter, we will learn how to calculate determinants for sizes greater than  $3 \times 3$ , and that determinants can be developed not only concerning the top row but also other rows or columns.

Additionally, we will familiarize ourselves with properties that apply to determinants, which can sometimes simplify the calculation of the determinant.

Finally, we will address the geometric interpretation of the determinant. However, let's start with an introduction that presents how the concept of a determinant arises quite naturally when studying linear  $n \times n$  systems of equations.

# Introduction to the Concept of Determinant

Matrices are a relatively recent concept in mathematics, as their study can be said to have begun only around the mid-19th century. In contrast, linear systems of equations have been studied for at least 4000 years. A central question concerning these systems is whether the system has a solution, and if it does, whether that solution is unique. For an  $n \times n$  system, determining this is quite straightforward, although labor-intensive. Next, we will demonstrate how this works for the case  $n = 2$ .

We aim to solve an arbitrary  $2 \times 2$  system of equations

$$\begin{cases} ax + by = e \\ cx + dy = f \end{cases}$$

and determine when it has a unique solution. The system can be solved in the traditional way by, for instance, solving for one of the variables and substituting it into the other equation.

From the first equation, we have

$$x = \frac{e - by}{a},$$

so we get

$$c \frac{e - by}{a} + dy = f$$

$$ce - bcy + ady = af$$

$$y = \frac{af - ce}{ad - bc}$$

and further

$$x = \frac{e - b \frac{af - ce}{ad - bc}}{a} = \frac{\frac{e(ad - bc)}{ad - bc} - b \frac{af - ce}{ad - bc}}{a} = \frac{\left( \frac{ade - abf}{ad - bc} \right)}{a} = \frac{ade - abf}{ad - bc} \frac{1}{a} = \frac{de - bf}{ad - bc}.$$

Thus, we found a unique solution for the system

$$\begin{cases} x = \frac{de - bf}{ad - bc} \\ y = \frac{af - ce}{ad - bc} \end{cases},$$

and it seems to exist as long as  $ad - bc \neq 0$ .

To be precise, and we are indeed precise now, it can be noted from the intermediate steps of the solution that the previous solution does not indicate anything for the case where  $a \neq 0$ , because in this case, we would be dividing by zero.

To examine this case, we need to consider the situation prior to division by zero, set  $a = 0$ , and handle the case obtained in this way separately. Next, we will show that in this case as well, the solution can be expressed in the form obtained earlier.

Let  $a = 0$ , resulting in the system of equations:

$$\begin{cases} by = e \\ cx + dy = f. \end{cases}$$

In this case, we get  $y = \frac{e}{b}$  and then

$$cx + d\frac{e}{b} = f$$

$$bcx + de = bf$$

$$x = \frac{bf - de}{bc}.$$

These solutions can also be expressed in the previous form:

$$x = \frac{bf - de}{bc} = \frac{bf - de}{bc - 0} = \frac{bf - de}{bc - ad} = \frac{de - bf}{ad - bc},$$

$$y = \frac{e}{b} = \frac{ce}{bc} = \frac{0 - ce}{0 - bc} = \frac{af - ce}{ad - bc}.$$

This solution still has the problem of not accounting for the cases  $b = 0$  and  $c = 0$ , where division by zero occurs in the intermediate steps. In the case where  $b = 0$ , the system of equations has either a multivalued solution or no solution, depending on the value of the constant  $e$ . In the case where  $c = 0$ , the solution doesn't depend on the variable  $x$ , so it is either multivalued or doesn't exist.

In summary, we can conclude that a  $2 \times 2$  system of equations has a unique solution if and only if  $ad - bc \neq 0$ . It is essential that this condition does not depend in any way on the “right side” of the system of equations but solely on the coefficients of the variables.

When the same reasoning is repeated for the  $3 \times 3$  case, it is observed that the system

$$\begin{cases} ax + by + cz = j \\ dx + ey + fz = k \\ gx + hy + iz = l \end{cases}$$

has a unique solution if and only if  $aei - afh - bdi + bfg + cdh - ceg \neq 0$ . The expression that determines the uniqueness of the solution again depends only on the coefficients on the “left side” of the system.

This idea generalizes similarly to an  $n \times n$  system of equations, meaning that the uniqueness of the solution always depends on the coefficients of the variables with a specific regularity of the computed expression. Since the value of this expression determines the uniqueness of the solution to the system, it is referred to as the **determinant**.

What then does the already observed notation for the determinant

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}, \quad (5)$$

refer to? The formulas for determinants are quite cumbersome to remember. In the general  $n \times n$  case, the expression for the determinant contains  $n!$  terms, so the number of terms grows alarmingly fast as the size of the system increases. Additionally, each term contains  $n$  coefficients that need to be chosen correctly. However, it has been noted that from the notation (5), the expressions can be calculated using quite easily memorable rules, which have already been encountered earlier. This notation is simply a shorthand notation for expressions of a certain form.



In summary, we can state that the determinant is fundamentally a number defined from the coefficients of an  $n \times n$  system according to a certain rule, which indicates whether the solution to the system is unique.

Moreover, it can be utilized, for example, when calculating the cross product, determining the inverse matrix, examining linear dependence, solving systems of equations, calculating the volume of a parallelepiped, determining the conversion ratio of a coordinate transformation, classifying critical points of a function of multiple variables, performing variable substitutions for integrating functions of several variables, and so on.

The applications of the determinant are thus numerous, but the same things could also be done without it, although the formulas would then take a more cumbersome form. The determinant is merely a calculation rule that allows a relatively complicated concept to sometimes be expressed in a fairly easy-to-remember and implementable form.

# Definition of the Determinant

Let's consider  $n \times n$  matrices, denoted by

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}. \quad (6)$$

In the case of  $n = 2$ , the determinant is defined by the formula

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}. \quad (7)$$

This formula for the determinant of a  $2 \times 2$  matrix is a basic formula that can be found in most formula collections. The formula for the determinant of a  $3 \times 3$  matrix is also often found in some form in these collections. For example, the following form, developed with respect to the top row, is widely used:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}. \quad (8)$$

The  $2 \times 2$  determinants on the right side of the equation, known as *minors*, are obtained by removing the corresponding row and column from the original determinant for that minor's coefficient. For example, in the case of the coefficient  $a_{11}$ , the first row and the first column are removed. Note the signs of the terms corresponding to the minors:  $+$ ,  $-$ , and  $+$ .

### Example

$$\begin{aligned}\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} &= 1 \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 2 \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3 \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} \\ &= 1(5 \cdot 9 - 6 \cdot 8) - 2(4 \cdot 9 - 6 \cdot 7) + 3(4 \cdot 8 - 5 \cdot 7) \\ &= -3 + 12 - 9 \\ &= 0\end{aligned}$$

## Basic Exercise

- (a) Calculate the determinant of the matrix

$$\begin{vmatrix} 3 & -1 & 4 \\ 1 & 3 & -1 \\ 0 & 2 & 0 \end{vmatrix}$$

- (b) Calculate the cross product of the vectors  $\mathbf{u} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$  and  $\mathbf{v} = -2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$  using the determinant.

The previously discussed  $3 \times 3$  case is a special case of a recursive formula that is convenient for defining the determinant of larger matrices:

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \sum_{k=1}^n (-1)^{j+k} a_{jk} M_{jk},$$

where the minor  $M_{jk}$  is the determinant of order  $n - 1$ , obtained by removing the  $j$ -th row and the  $k$ -th column from the matrix. The index  $j$  can thus refer to any row. The determinant calculated in this way is said to be expanded with respect to the row.

The determinant can also be expanded with respect to a column:

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \sum_{j=1}^n (-1)^{j+k} a_{jk} M_{jk},$$

the final result is the same.

Note that the signs of the coefficients  $a_{jk}$ ,  $(-1)^{j+k}$ , do not need to be calculated separately, as they repeat regularly in the following manner:

$$\begin{vmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & + & - & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{vmatrix}.$$

Often, the term used for the sign of the cofactor and its corresponding minor is called the *cofactor*: the cofactor of  $a_{jk}$  is defined as

$C_{jk} = (-1)^{j+k} M_{jk}$ , and the determinant can be expressed in terms of cofactors as:

$$\det(A) = a_{j1}C_{j1} + a_{j2}C_{j2} + \cdots + a_{jn}C_{jn} = a_{1k}C_{1k} + a_{2k}C_{2k} + \cdots + a_{nk}C_{nk}. \quad (9)$$

The determinant can therefore be expanded with respect to any row or column. Generally, it is most reasonable to choose the row or column with the most zeros.

### Example

Determine  $\det(A)$  when

$$A = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -1 & 0 & 0 & 2 \\ 2 & 1 & 3 & 0 \\ 0 & 4 & 0 & 2 \end{bmatrix}.$$

### Solution

For the sake of the example, we will expand the matrix with respect to both the bottom row and the third column.



### Solution (continued)

With respect to the bottom row, we quickly get

$$\begin{aligned}\det(A) &= \begin{vmatrix} 0 & 1 & 0 & -1 \\ -1 & 0 & 0 & 2 \\ 2 & 1 & 3 & 0 \\ \mathbf{0} & \mathbf{4} & \mathbf{0} & \mathbf{2} \end{vmatrix} \\ &= -0 + 4 \cdot \begin{vmatrix} \mathbf{0} & \mathbf{0} & \mathbf{-1} \\ -1 & 0 & 2 \\ 2 & 3 & 0 \end{vmatrix} - 0 + 2 \cdot \begin{vmatrix} \mathbf{0} & \mathbf{1} & \mathbf{0} \\ -1 & 0 & 0 \\ 2 & 1 & 3 \end{vmatrix} \\ &= 4(0 - 0 - 1(-3 - 0)) + 2(0 - 1(-3 - 0) + 0) \\ &= 12 + 6 \\ &= 18.\end{aligned}$$

### Solution (continued)

Here is a version that uses the formula with more intermediate steps:

$$\begin{aligned}\det(A) &= \begin{vmatrix} 0 & 1 & 0 & -1 \\ -1 & 0 & 0 & 2 \\ 2 & 1 & 3 & 0 \\ \mathbf{0} & \mathbf{4} & \mathbf{0} & \mathbf{2} \end{vmatrix} = (-1)^{4+1} \cdot 0 \cdot \begin{vmatrix} 1 & 0 & -1 \\ 0 & 0 & 2 \\ 1 & 3 & 0 \end{vmatrix} + (-1)^{4+2} \cdot 4 \cdot \begin{vmatrix} 0 & 0 & -1 \\ -1 & 0 & 2 \\ 2 & 3 & 0 \end{vmatrix} \\ &+ (-1)^{4+3} \cdot 0 \cdot \begin{vmatrix} 0 & 1 & -1 \\ -1 & 0 & 2 \\ 2 & 1 & 0 \end{vmatrix} + (-1)^{4+4} \cdot 2 \cdot \begin{vmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 2 & 1 & 3 \end{vmatrix} \\ &= \underbrace{(-1)^6}_{1} \cdot 4 \cdot \begin{vmatrix} 0 & \mathbf{0} & -1 \\ -1 & \mathbf{0} & 2 \\ 2 & \mathbf{3} & 0 \end{vmatrix} + \underbrace{(-1)^8}_{1} \cdot 2 \cdot \begin{vmatrix} \mathbf{0} & \mathbf{1} & \mathbf{0} \\ -1 & 0 & 0 \\ 2 & 1 & 3 \end{vmatrix} \\ &= 4 \cdot \left( (-1)^{1+2} \cdot 0 \begin{vmatrix} -1 & 2 \\ 2 & 0 \end{vmatrix} + (-1)^{2+2} \cdot 0 \begin{vmatrix} 0 & -1 \\ 2 & 0 \end{vmatrix} + (-1)^{3+2} \cdot 3 \begin{vmatrix} 0 & -1 \\ -1 & 2 \end{vmatrix} \right) \\ &+ 2 \cdot \left( (-1)^{1+1} \cdot 0 \begin{vmatrix} 0 & 0 \\ 1 & 3 \end{vmatrix} + (-1)^{1+2} \cdot 1 \begin{vmatrix} -1 & 0 \\ 2 & 3 \end{vmatrix} + (-1)^{1+3} \cdot 0 \begin{vmatrix} -1 & 0 \\ 2 & 1 \end{vmatrix} \right) \\ &= 4 \cdot (-3) \begin{vmatrix} 0 & -1 \\ -1 & 2 \end{vmatrix} + 2 \cdot (-1) \begin{vmatrix} -1 & 0 \\ 2 & 3 \end{vmatrix} \\ &= -12(0 \cdot 2 - (-1) \cdot (-1)) - 2(-1 \cdot 3 - 2 \cdot 0) \\ &= -12 \cdot (-1) - 2 \cdot (-3) = 18.\end{aligned}$$

### Solution (continued)

Expanding with respect to column 3 yields the simplest calculation (Why?):

$$\det(A) = \begin{vmatrix} 0 & 1 & \mathbf{0} & -1 \\ -1 & 0 & \mathbf{0} & 2 \\ 2 & 1 & \mathbf{3} & 0 \\ 0 & 4 & \mathbf{0} & 2 \end{vmatrix} = 3 \begin{vmatrix} \mathbf{0} & 1 & -1 \\ -\mathbf{1} & 0 & 2 \\ \mathbf{0} & 4 & 2 \end{vmatrix} = 3(0 - (-1)(2+4) + 0) = 18.$$

### Basic Exercise

Calculate the determinant of the matrix

$$\begin{bmatrix} 3 & -1 & 4 \\ 1 & 3 & -1 \\ 0 & 2 & 0 \end{bmatrix}$$

- (a) with respect to the first row
- (b) with respect to the easiest row/column.

# Basic Properties of Determinants

Perhaps the most fundamental property of the determinant relates to the existence of the inverse matrix and thus the unambiguity of the solution to an  $n \times n$  system of equations:

A matrix  $A$  has an inverse matrix if and only if  $\det(A) \neq 0$ .

The reason for this can be seen in the formula for the inverse matrix presented in the next section, where division by zero occurs if the determinant is zero.

The determinant also has several other properties, not all of which are immediately obvious. Here are some of them. The mentioned matrices are always square matrices, as determinants are not defined for other types of matrices.

- (1) If the rows and columns of a determinant are interchanged, the value of the determinant does not change, i.e.,  $\det(A^T) = \det(A)$ .
- (2) If a row or column of the determinant is multiplied by a constant  $k$ , the value of the determinant is multiplied by  $k$ . In the case of  $n \times n$ , it holds that  $\det(kA) = k^n \det(A)$ .
- (3) If all elements of a row or column are zeros, the value of the determinant is zero.
- (4) If a row or column can be obtained from another by multiplying it by a constant, the value of the determinant is zero.
- (5) If a row or column is added to another multiplied by a constant, the value of the determinant does not change.
- (6) If two rows or columns of the determinant are interchanged, the sign of the determinant changes.
- (7) The determinant of lower and upper triangular matrices, and thus also of diagonal matrices, is given by the product of the diagonal elements.

(8) For the product of matrices:

$$\det(AB) = \det(BA) = \det(A) \det(B).$$

(9) For the inverse matrix:

$$\det(A^{-1}) = \det(A)^{-1} = \frac{1}{\det(A)}.$$

- (10) An  $n \times n$  system of equations has a unique solution if and only if the determinant of the coefficient matrix is not zero.
- (11) The trivial solution  $\mathbf{x} = \mathbf{0}$  of a homogeneous  $n \times n$  system of equations is the only solution if and only if the determinant of the coefficient matrix is not zero.
- (12) The rows (or columns) of a square matrix are linearly independent if and only if the determinant of the matrix is not zero.

From properties (5) to (7), it follows that the determinant can be computed by first reducing the matrix to row echelon form using Gaussian elimination and then multiplying the diagonal elements together. However, it is necessary to keep track of how many times rows or columns have been swapped to obtain the correct sign of the determinant.

### Example

Determine the determinant of the matrix

$$\begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 2 & 1 & 3 \end{bmatrix}$$

using Gaussian algorithm.

## Solution

$$\begin{vmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 2 & 1 & 3 \end{vmatrix} \begin{matrix} \\ +R_1 \\ -2R_1 \end{matrix}$$

$$= \begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 1 & 5 \end{vmatrix} -R_2$$

$$= \begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 6 \end{vmatrix}$$

$$= 1 \cdot 1 \cdot 6$$

$$= 6$$



## Basic Exercise

Calculate the determinants of the following matrices:

$$(a) \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

$$(b) \begin{bmatrix} 2 & 4 & 4 \\ 3 & -6 & 8 \\ 2 & 4 & -7 \end{bmatrix} \begin{bmatrix} 2 & 9 & 0 \\ -9 & 4 & 0 \\ -2 & 1 & 0 \end{bmatrix}$$

$$(c) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 10 \end{bmatrix}.$$

## MATLAB Example

The determinant of the matrix

$$\begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 2 & 1 & 3 \end{bmatrix}$$

is calculated in MATLAB as follows:

```
A = [1, 0, -1; -1, 1, 0; 2, 1, 3];  
d = det(A)
```